EE6301
Review and Source Comparison
复核报告与重点对比

Evidence, priority decisions and corrections
依据、筛选决定与纠错

Body text / 正文：11 pt
English first, Chinese immediately below. / 英文在前，中文紧随。

2026-10-11 · Course notes, sample explanations and original practice
课件知识、样题参考解析与原创练习

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Review conclusions / 本次复核结论

EN: The earlier guide was broadly correct but too narrow in recognition coverage. This review retains 25 main cards, replaces six broad secondary paragraphs with 32 concise entries, and keeps the mock at 25 questions. A priority label is not a prediction probability. The lecturer explicitly warns that the samples are only examples (T8-0060).

中文： 上一版主要原理总体正确，但概念辨识覆盖偏窄。本轮保留25张主卡，将六段粗略补充替换为32条简短知识，并保持25题模拟规模。优先级不等于出题概率；老师明确说样题只是示例（T8-0060）。

EN: The 157 source-topic records were each assigned a location or an explicit reason for lower emphasis. These are indexing groups, not 157 separate examinable facts. The CSV preserves all decisions. Several rows cover both basic concepts and detailed calculations; a C label applies to depth, not an automatic topic-wide exclusion.

中文： 已逐项为157条来源主题分配学习位置或降级理由。这些是索引主题组，不是157个独立必考知识点。CSV保留全部处理决定。多条记录同时包含基础概念和详细计算；C针对深度，并非自动把整个主题排除。

Restored breadth by week / 各周补回内容

EN: W1: recognition elements, sensor families, linearity/response time/reproducibility, POCT and microfluidics.

中文： W1：识别元件、传感器类别、线性度/响应时间/再现性、POCT与微流控。

EN: W2: light sources/coherence, scattering and attenuation, pulse oximetry, phosphorescence/state transitions and Raman-shift conversion.

中文： W2：光源/相干性、散射与衰减、血氧、磷光/能级跃迁、拉曼位移换算。

EN: W3: quantum dots, FSC/SSC/fluorescence in flow cytometry, fibre modes/cavity types, Q/FWHM/FOM and SERS.

中文： W3：量子点、流式FSC/SSC/荧光、光纤模式/腔型、Q/FWHM/FOM、SERS。

EN: W4: imaging modalities, optical biopsy/virtual staining, NA versus sampling, PMT, OCT axial/lateral/depth-of-focus distinctions.

中文： W4：成像技术、光学活检/虚拟染色、NA与采样、PMT、OCT轴向/横向/焦深区别。

EN: W5: piezoelectric transmit/receive, acoustic impedance/attenuation, photoacoustic contrast, X-ray generation and basic image processing.

中文： W5：压电收发、声阻抗/衰减、光声对比、X射线产生与基本图像处理。

EN: W6: redox, microfabrication/PDMS, three glucose generations, Faraday charge relations, FET/nanopore/capacitance and Bode meaning.

中文： W6：氧化还原、微加工/PDMS、三代葡萄糖传感、法拉第电荷关系、FET/纳米孔/电容及Bode含义。

EN: W7: all-or-none firing, ionic gradients/Nernst/GHK, the electrode interface, ECG leads/heart rate and EEG bands.

中文： W7：全或无发放、离子梯度/Nernst/GHK、电极界面、心电导联/心率、脑波类别。

EN: W8: amplification stages, isolation, impedance and inverting/non-inverting recognition; the scope still ends at p29.

中文： W8：放大链、隔离、阻抗、反相/同相辨识；范围仍止于p29。

Factual corrections and boundaries / 事实纠错与边界

EN: PMT correction: the lecture description at T4-0049–0050 wrongly describes tiny mirrors and optical reflection. The actual diagram (W4 p35) labels dynodes: multiplication is by secondary electron emission. Amplification does not remove all noise. See Hamamatsu’ s PMT explanation.

中文： PMT纠错：T4-0049–0050把结构说成小镜子与光反射，讲解不准确。原图W4 p35标有dynodes倍增极，实际为二次电子发射倍增；放大不等于消除全部噪声。已核对 Hamamatsu官方PMT说明。

EN: The X-ray “non-ionizing” error also occurs in the original W4 p7, not only a student sheet. Correct it to ionizing; see FDA medical X-ray imaging. W7 p84 also misprints gamma as <25 Hz; learn the band ordering here rather than memorizing this inconsistent boundary.

中文： X射线“non-ionizing”的错误也出现在原课件W4 p7，不只是学生一页纸；应改为电离性，见FDA医学X射线说明。W7 p84还把gamma误写为<25 Hz；这里保留频带顺序，不让你死背不一致的边界。

EN: Further wording safeguards: the glucose/peroxide example is not universal to amperometry; high input impedance is not voltage gain; below-threshold neurons still have a resting potential; ECG leads are not the P/QRS/T waves. These conceptual corrections take precedence over loose oral wording.

中文： 另已防止四种过度概括：葡萄糖/过氧化氢例子不代表全部安培传感；高输入阻抗不等于高电压增益；未达阈值仍有静息电位；心电导联不等于P/QRS/T波。复习采用纠正后的概念，不照抄宽泛口头表述。

EN: Additional precision: LOD improves with increasing slope only if noise is held fixed; a spectral FOM must use matching wavelength or frequency coordinates. The original W3 p80 mixes labels, so the reviewed guide writes the wavelength form explicitly. Definitions of Q/FOM were restored even though long final-style calculations remain secondary.

中文： 另作两点精确化：斜率增加改善LOD需固定噪声；光谱FOM的灵敏度与线宽必须匹配波长/频率坐标。原W3 p80标注混用，复核版明确写波长形式。Q/FOM定义已补回，长的期末型计算仍次要。

EN: Preserved lower-priority boundaries: detailed historical chronology; research/vendor case specifics; long SPR, OCT, multi-ion and complex-impedance derivations; detailed ultrasound resolution calculations; the locally deemphasized RC example. Week 8 p30 onward and the conformal-electrode design requirements remain outside this review scope. This does not exclude basic ultrasound, OCT, ion-gradient or filter concepts.

中文： 保留的低优先级边界：历史年份、论文/厂商案例细节、长SPR/OCT/多离子/复阻抗推导、详细超声分辨率计算及局部弱化的RC例题。Week8 p30起与贴合电极设计要求仍不纳入本次范围；这不排除基础超声、OCT、离子梯度或滤波概念。

Quiz verification / 模拟卷核验

EN: All 25 English/Chinese stems and 125 paired options were reviewed for meaning and a single best answer. No answer-letter change was needed. The eight calculation answers were recomputed independently: Q6 6.25%; Q7 0.050 ns^{-1}; Q8 5000; Q9 0.45 mM; Q12 3.08 cm; Q23 0.20 V; Q24 2.0 μA; Q25 5 viruses.

中文： 已逐一复核25组英中文题干与125组对应选项的含义及唯一最佳答案，无需改动答案字母。独立重算八道计算：Q6 6.25%；Q7 0.050 ns^{-1}；Q8 5000；Q9 0.45 mM；Q12 3.08 cm；Q23 0.20 V；Q24 2.0 μA；Q25 5个病毒。

EN: The mock includes necessary assumptions: equal illuminated grating width/order, adequate detector sampling, identical common-mode noise, a simplified pure-resistor circuit, and calibration validity near the integer detection threshold. It is an introductory diagnostic exercise, not a complete topic inventory or a prediction of exact questions.

中文： 模拟题已写明必要假设：光栅同受照宽度/级次、充分探测器采样、完全相同共模噪声、纯电阻简化电路，以及整数检测阈值附近校准斜率有效。它用于基础诊断，不是完整知识目录，也不预测真实题目原样出现。

EN: Spring sample references after review: Q1 D; Q2 E under the course’ s broad resonator classification; Q3 A under its labelled-fluorescence convention; Q4 A; Q5 C provisionally only; Q6 A; Q7 A (second order as printed); Q8 A; Q9 B by detector principles; Q10 B=5; Q11 B; Q12 A as intended arithmetic. Q5 lacks a robust official numeric resolution, Q9 lacks a clear spoken final letter, and Q12 has incompatible current/current-density units.

中文： 春季样题复核参考：Q1 D；Q2 E（按课程广义共振器分类）；Q3 A（按课程有标记荧光分类）；Q4 A；Q5仅暂定C；Q6 A；Q7 A（按原题第二级）；Q8 A；Q9依据探测器原理选B；Q10 B=5；Q11 B；Q12题意算术选A。Q5无法可靠确定官方数值，Q9口头未明确最后字母，Q12存在电流/电流密度单位冲突。

Evidence and review limits / 证据与复核范围

EN: This review used the existing full transcript index and 157-topic register, reread relevant slide pages and transcript passages, inspected the original three sample pages, and checked selected diagrams including PMT, glucose generations and EEG bands. It did not newly rewatch every lecture video or fully solve every historical final problem. OCR and ASR remain retrieval aids, not authoritative mathematical sources.

中文： 本轮使用既有完整字幕索引与157主题记录，回读相关课件页和字幕，查看原样卷全部三页，并核对PMT、葡萄糖三代与脑波等选定图示。未在本轮重新观看全部讲课视频或逐题解完历年期末。OCR和字幕识别仅作检索辅助，不当作公式权威。

EN: The comparison with past final sheets is retained below/in this report. Dense or repeated final-sheet content is not evidence of an exact Quiz probability. The separate coverage CSV and arithmetic JSON support checking the changes without enlarging the main study guide.

中文： 原有课堂与往年期末一页纸对比保留在本报告。期末笔记写得密或重复出现，不能当作精确Quiz概率。独立覆盖CSV和算术JSON用于核验改动，不挤占主复习正文。

EN: The decision rule is: current scope and explicit Quiz statements → current worked sample → repeated foundational teaching → corroborating final-sheet overlap. Past final recurrence supports familiarity, but cannot establish a probability for this Quiz or turn a final-exam promise into a Quiz promise.

中文： 决策顺序：本次范围与明确Quiz声明→当前样题讲解→反复讲解的基础→历年期末重合。期末重复支持熟悉度，但不能据此计算本次出题概率，也不能把期末承诺改说成Quiz承诺。

Source inventory and limits / 来源清单与分析边界

EN: Eight transcripts: 124,232 words, 831 indexed segments. All segments were reviewed across three assignments, yielding 157 topic records with source quotes, assessment targets and ASR notes. The 87-page final compilation was extracted/OCRed and viewed across all pages at overview scale; relevant anchors were checked more closely. This is a priority comparison, not a complete answer-key audit of every historical problem.

中文： 八份字幕共124,232词、831个索引段；三组审读覆盖全部段，形成157条主题记录，含原文引句、考核指向与字幕纠错。87页期末合集已提取/OCR并逐页概览，相关重点位置进一步核对。本次做重点对比，并非逐题完整核验全部历史试卷答案。

EN: past1 = Cheatsheet+Final合集+答案.pdf (87 pages): sheets, exam papers, student/annotated solutions and repeated answer material. past2 = paper (1).pdf (2 pages): typed mixed-semester notes with image formulas. past3 = 期末一页纸.pdf (2 pages): scanned illustrated notes. jpg1/jpg2 are the two WeChat images. These are not five statistically independent past exams.

中文： past1=Cheatsheet+Final合集+答案.pdf（87页）：混合一页纸、试卷、学生/批注解答及重复答案材料；past2=paper (1).pdf（2页）：打字笔记与公式图，跨课程段；past3=期末一页纸.pdf（2页）：扫描图文笔记；jpg1/jpg2为两张微信图片。这些不能当作五场独立历史考试来统计。

EN: The compilation contains identifiable April/May 2023, November 2023 and November 2024 final papers and solutions. Later reproduced solution pages overlap earlier questions. Printed “solutions” or “secret” labels do not independently verify official provenance. Formula OCR is used for retrieval, then checked against images or course content; Greek symbols and denominators are not trusted blindly.

中文： 合集包含可辨识的2023年4/5月、2023年11月及2024年11月期末试卷与解答；后段复现的答案页与前面题目重合。印有“solutions”或“secret”字样不能独立证明官方来源。公式OCR用于检索，再按图或课程内容核对；不能直接相信识别出的希腊字母与分母。

Topic-by-topic decisions / 逐主题筛选决定

[A] LOD and calibration / LOD与校准

EN: Past overlap: past1 p3, p20–21, p64, p71; past2 p1; past3 p1–2

> **中文：** 历年重合：past1 p3, p20–21, p64, p71; past2 p1; past3 p1–2

EN: Current classroom evidence: W1 PDF p60–62; Sample quiz p2 Q10; W1 p59–62 | T1-0057–0058, T7-0057–0060

> **中文：** 本年课堂依据：W1 PDF p60–62; Sample quiz p2 Q10; W1 p59–62 | T1-0057–0058, T7-0057–0060

EN: Decision: Repeated final presence agrees with an explicit Quiz/final promise and a worked Quiz sample. Keep slope/noise/linear-region logic and the integer threshold correction.

> **中文：** 筛选决定：历年出现与老师明确Quiz/期末声明及样题一致。保留斜率、噪声、线性区逻辑及整数阈值纠错。

[A] Beer–Lambert / 比尔—朗伯

EN: Past overlap: past1 p3, p9, p71; past2 p1; past3 p2

> **中文：** 历年重合：past1 p3, p9, p71; past2 p1; past3 p2

EN: Current classroom evidence: W2 PDF p36–40; W2 PDF p41–43; Sample quiz p2 Q11; Sample quiz p3 Q12; W2 p37–43; W1 p61–62; W8 classroom announcements; W2 p37–43 (named A=εCL example) | T2-0041–0051, T7-0060–0061, T8-0024, T8-0061–0064

> **中文：** 本年课堂依据：W2 PDF p36–40; W2 PDF p41–43; Sample quiz p2 Q11; Sample quiz p3 Q12; W2 p37–43; W1 p61–62; W8 classroom announcements; W2 p37–43 (named A=εCL example) | T2-0041–0051, T7-0060–0061, T8-0024, T8-0061–0064

EN: Decision: Keep the short A=εCL calculation; the lecturer specifically names it as a memory target.

> **中文：** 筛选决定：保留短式A=εCL计算；老师明确用它举例说明需要记忆。

[A] Fluorescence, yield and lifetime / 荧光、产率与寿命

EN: Past overlap: past1 p42, p50; past2 p1; past3 p1–2

> **中文：** 历年重合：past1 p42, p50; past2 p1; past3 p1–2

EN: Current classroom evidence: W2 PDF p69–76; W2 PDF p77–79; W2 PDF p82–88; Sample quiz p1 Q1–3; W2 p73–78; W3 p39–50; W4 p54–61; Sample quiz p1 Q6; Sample quiz p2 Q7; W2 p82–88 | T2-0088–0100, T2-0105–0107, T7-0051–0055

> **中文：** 本年课堂依据：W2 PDF p69–76; W2 PDF p77–79; W2 PDF p82–88; Sample quiz p1 Q1–3; W2 p73–78; W3 p39–50; W4 p54–61; Sample quiz p1 Q6; Sample quiz p2 Q7; W2 p82–88 | T2-0088–0100, T2-0105–0107, T7-0051–0055

EN: Decision: Keep mechanisms, filters and Φ/τ arithmetic; complicated photon-budget examples target the final more directly.

> **中文：** 筛选决定：保留机制、滤光与Φ/τ短算；复杂光子预算例题更直接针对期末。

[A] Grating density and resolving power / 光栅密度与分辨本领

EN: Past overlap: past1 p3, p9, p43, p53; past2 p1; past3 p2

> **中文：** 历年重合：past1 p3, p9, p43, p53; past2 p1; past3 p2

EN: Current classroom evidence: W2 p35 (spectrometer structure background; no 300/1200 comparison on supplied slides); Sample quiz p1 Q6; Sample quiz p2 Q7; W2 p82–88 | T5-0061–0063, T7-0054–0055

> **中文：** 本年课堂依据：W2 p35 (spectrometer structure background; no 300/1200 comparison on supplied slides); Sample quiz p1 Q6; Sample quiz p2 Q7; W2 p82–88 | T5-0061–0063, T7-0054–0055

EN: Decision: The explicit Week 5 Quiz promise raises the density comparison. Do not invent a Week 2 slide for 300/1200 lines/mm.

> **中文：** 筛选决定：Week5明确Quiz承诺使密度比较上调；不能虚标300/1200图的Week2页码。

[A] NA and magnification / NA与放大倍率

EN: Past overlap: past1 p12, p41, p47–49, p70; past2 p1; past3 p2

> **中文：** 历年重合：past1 p12, p41, p47–49, p70; past2 p1; past3 p2

EN: Current classroom evidence: W4 p27–30; W4 p29–34; W4 p29–30 | T4-0030–0038, T5-0059–0061

> **中文：** 本年课堂依据：W4 p27–30; W4 p29–34; W4 p29–30 | T4-0030–0038, T5-0059–0061

EN: Decision: An earlier “every year” statement targets the final; the later Week 5 statement separately confirms a Quiz comparison.

> **中文：** 筛选决定：较早“每年考”针对期末；Week5后续另有明确Quiz比较承诺，二者需分开。

[A/B] Label-free, fibers and resonators / 无标记、光纤与共振

EN: Past overlap: past1 p4, p10–11, p21, p27; past2 p1; past3 p1–2

> **中文：** 历年重合：past1 p4, p10–11, p21, p27; past2 p1; past3 p1–2

EN: Current classroom evidence: W3 PDF p13–18; W3 PDF p19–24; W3 PDF p39–45; W3 PDF p51–60; W3 PDF p64–75; Sample quiz p1 Q1–3; W2 p73–78; W3 p39–50; W4 p54–61 | T3-0016–0020, T3-0030–0038, T3-0062–0072, T3-0084–0093, T3-0099–0105, T7-0051–0053

> **中文：** 本年课堂依据：W3 PDF p13–18; W3 PDF p19–24; W3 PDF p39–45; W3 PDF p51–60; W3 PDF p64–75; Sample quiz p1 Q1–3; W2 p73–78; W3 p39–50; W4 p54–61 | T3-0016–0020, T3-0030–0038, T3-0062–0072, T3-0084–0093, T3-0099–0105, T7-0051–0053

EN: Decision: Keep recognition and resonance principles. Long calibration, linewidth/FOM and angle-interrogation calculations are secondary or final-focused.

> **中文：** 筛选决定：保留分类与共振原理；长校准、线宽/FOM和角度探询计算次要或偏期末。

[A] ART and LSPR / 快检与LSPR

EN: Past overlap: past3 p1 explicitly labels ART; past2 p1 includes LSPR

> **中文：** 历年重合：past3 p1 explicitly labels ART; past2 p1 includes LSPR

EN: Current classroom evidence: W3 PDF p76; W7 p52–55 (context; classroom question diagram); W3 p76–79 | T3-0106, T7-0047–0050

> **中文：** 本年课堂依据：W3 PDF p76; W7 p52–55 (context; classroom question diagram); W3 p76–79 | T3-0106, T7-0047–0050

EN: Decision: The current lecturer explicitly says this is a recurring Quiz question. It outranks a topic that appears densely in a final sheet without a Quiz promise.

> **中文：** 筛选决定：本年老师明确说这是反复出的Quiz题，优先于仅在期末笔记中密集出现却无Quiz承诺的内容。

[A core / C derivation] OCT / 光学相干断层

EN: Past overlap: past1 p5, p23, p69, p78; past2 p1; past3 p2

> **中文：** 历年重合：past1 p5, p23, p69, p78; past2 p1; past3 p2

EN: Current classroom evidence: W4 p71–79; W4 p72–79; W4 p80–82; W4 p85–90; Sample quiz p2 Q8–9; W4 p35–38, p73–86; W3 p25–32 | T4-0088–0102, T4-0105–0108, T7-0055–0057

> **中文：** 本年课堂依据：W4 p71–79; W4 p72–79; W4 p80–82; W4 p85–90; Sample quiz p2 Q8–9; W4 p35–38, p73–86; W3 p25–32 | T4-0088–0102, T4-0105–0108, T7-0055–0057

EN: Decision: Sample Q8 supports mechanism and reflectivity. Lower long interference algebra under the global short-calculation guidance; no topic-specific blanket exclusion was found.

中文： 筛选决定：样题Q8支持机制与反射率；依据总的短计算要求降低长干涉代数，未找到主题层面的全盘排除。

[B core / C detailed resolution] Ultrasound and photoacoustics / 超声与光声

EN: Past overlap: past1 p6, p13, p23–24, p45, p57, p70; past2 p1; past3 p1–2

中文： 历年重合：past1 p6, p13, p23–24, p45, p57, p70; past2 p1; past3 p1–2

EN: Current classroom evidence: W5 p5–14; W5 p27–28; W5 p29; W5 p34–43 | T5-0004–0011, T5-0037–0040, T5-0044–0051

中文： 本年课堂依据：W5 p5–14; W5 p27–28; W5 p29; W5 p34–43 | T5-0004–0011, T5-0037–0040, T5-0044–0051

EN: Decision: Retain signal chains and simple timing. The technical scan/resolution passages are locally deemphasized despite detailed final calculations.

中文： 筛选决定：保留信号链与简单时间关系；虽然期末有详细计算，课堂仍对技术扫描与分辨率作局部弱化。

[A/B] CCD/CMOS and HSI / CCD/CMOS与高光谱

EN: Past overlap: past1 p25, p38, p85; past2 p1; past3 p1

中文： 历年重合：past1 p25, p38, p85; past2 p1; past3 p1

EN: Current classroom evidence: W4 p35–38; W4 p91–104; W4 p105; Sample quiz p2 Q8–9; W4 p35–38, p73–86; W3 p25–32 | T4-0048–0055, T4-0109–0114, T7-0055–0057

中文： 本年课堂依据：W4 p35–38; W4 p91–104; W4 p105; Sample quiz p2 Q8–9; W4 p35–38, p73–86; W3 p25–32 | T4-0048–0055, T4-0109–0114, T7-0055–0057

EN: Decision: Keep detector distinctions and data-cube meaning. Vendor descriptions and every acquisition implementation are not equal-priority memorization targets.

中文： 筛选决定：保留探测器区别与数据立方含义；厂商介绍和所有采集实现并不都需同等记忆。

[B core / C complex stoichiometry] Electrochemical measurement / 电化学测量

EN: Past overlap: past1 p6, p14, p75; past2 p1; past3 p1–2

中文： 历年重合：past1 p6, p14, p75; past2 p1; past3 p1–2

EN: Current classroom evidence: W6 p19–21; W6 p37–40; W6 p41–46; W6 p50–52; W6 p60; W6 p55, p57–59 | T6-0019–0022, T6-0038–0043, T6-0054–0058, T6-0060–0073

中文： 本年课堂依据：W6 p19–21; W6 p37–40; W6 p41–46; W6 p50–52; W6 p60; W6 p55, p57–59 | T6-0019–0022, T6-0038–0043, T6-0054–0058, T6-0060–0073

EN: Decision: Keep potential/current/conductance/impedance classifications and electrode roles. Lactate mass-to-electron work is an exam-style signal with ambiguous Quiz targeting; lower it rather than calling it explicitly excluded.

> **中文：** 筛选决定：保留电位、电流、电导、阻抗分类及电极作用；乳酸质量→电子例题指向考试，但Quiz指向不明确，因此降低优先级而非称明确不考。

[A/B] Biopotentials and EEG/ECG / 生物电与脑电/心电

EN: Past overlap: past1 p7, p16–17, p26, p61, p73–74, p87; past2 p2; past3 p1–2

> **中文：** 历年重合：past1 p7, p16–17, p26, p61, p73–74, p87; past2 p2; past3 p1–2

EN: Current classroom evidence: W7 p5–8; W7 p19–21; W7 p32–39; Sample quiz p1 Q4; W1 p16, p48, p69–75; W7 p62–69; W7 p79–87; Sample quiz p1 Q5 | T7-0002–0006, T7-0011–0012, T7-0023–0025, T7-0028–0030, T7-0053, T7-0068–0073, T7-0084–0087, T7-0091–0095

> **中文：** 本年课堂依据：W7 p5–8; W7 p19–21; W7 p32–39; Sample quiz p1 Q4; W1 p16, p48, p69–75; W7 p62–69; W7 p79–87; Sample quiz p1 Q5 | T7-0002–0006, T7-0011–0012, T7-0023–0025, T7-0028–0030, T7-0053, T7-0068–0073, T7-0084–0087, T7-0091–0095

EN: Decision: Keep sources, phases and graph reading. Deemphasis of EMG numerical ranges is narrow; it does not remove basic EMG concepts.

> **中文：** 筛选决定：保留组织来源、动作电位阶段与读图。EMG数值范围的弱化是局部的，不排除基本EMG概念。

[A] Patch clamp and MEA / 膜片钳与多电极阵列

EN: Past overlap: past3 p1; jpg2

> **中文：** 历年重合：past3 p1; jpg2

EN: Current classroom evidence: W8 p9–13; W8 p9–10; W8 p11–13 | T8-0005–0016, T8-0018

> **中文：** 本年课堂依据：W8 p9–13; W8 p9–10; W8 p11–13 | T8-0005–0016, T8-0018

EN: Decision: The Week 8 lecturer explicitly names both methods. Keep diagram recognition and their key measurement difference.

> **中文：** 筛选决定：Week8明确点名两种方法；保留图示辨识和关键测量差异。

[B core / C local cutoff example] Amplifiers and filters / 放大器与滤波器

EN: Past overlap: past2 p2; past3 p2

> **中文：** 历年重合：past2 p2; past3 p2

EN: Current classroom evidence: W8 p19–20; W8 p20; W8 p22; W8 p25–26; W8 p27–28; W8 p29 | T8-0021–0029, T8-0062–0063

> **中文：** 本年课堂依据：W8 p19–20; W8 p20; W8 p22; W8 p25–26; W8 p27–28; W8 p29 | T8-0021–0029, T8-0062–0063

EN: Decision: Keep gain, input impedance, common-mode and filter roles. The specific RC example is locally excluded from this year’ s exam, while its exact Quiz implication remains qualified.

> **中文：** 筛选决定：保留增益、输入阻抗、共模与滤波作用；特定RC例题在局部语境中被排除于今年考试，其对Quiz的具体影响仍保留限定。

[C] Complex impedance and Bode / 复杂阻抗与Bode

EN: Past overlap: past1 p22, p30–34, p44, p54–55, p62, p81–84; past3 p2

> **中文：** 历年重合：past1 p22, p30–34, p44, p54–55, p62, p81–84; past3 p2

EN: Current classroom evidence: W6 p84–86; W6 p86; W7 p70–78 | T6-0096–0101, T7-0073–0081

> **中文：** 本年课堂依据：W6 p84–86; W6 p86; W7 p70–78 | T6-0096–0101, T7-0073–0081

EN: Decision: Dense final coverage agrees with explicit final-focused teaching. For the Quiz, keep V/I and qualitative reactance trends before long AC algebra.

> **中文：** 筛选决定：期末密集覆盖与明确针对期末的教学相符；Quiz先保留V/I和定性电抗趋势，长交流代数后置。

[C] Confusion matrix, ROC and long ML work / 混淆矩阵、ROC与长机器学习计算

EN: Past overlap: past1 p8, p18–19, p25–26, p46, p58–59, p76, p85–87; past3 p2

> **中文：** 历年重合：past1 p8, p18–19, p25–26, p46, p58–59, p76, p85–87; past3 p2

EN: Current classroom evidence: W5 p84–86 | T5-0104–0106

> **中文：** 本年课堂依据：W5 p84–86 | T5-0104–0106

EN: Decision: The current lecture calls the matrix/ROC material reference or not very important. Past-sheet density does not override this local signal.

> **中文：** 筛选决定：本年课堂将矩阵/ROC称为参考或不太重要；过去笔记的密度不能覆盖该局部信号。

[D for this scope] HMI, self-powered devices and later electronics / 人机交互、自供能与后续电子系统

EN: Past overlap: past2 p2; past3 p1–2; jpg2

> **中文：** 历年重合：past2 p2; past3 p1–2; jpg2

EN: Current classroom evidence: W8 p5–29 included; W8 p30–73 excluded | T8-0030, T8-0035–0037, T8-0060, T8-0065–0067, T8-0081–0084

> **中文：** 本年课堂依据：W8 p5–29 included; W8 p30–73 excluded | T8-0030, T8-0035–0037, T8-0060, T8-0065–0067, T8-0081–0084

EN: Decision: Week 8 p30 onward is outside the stated scope. Later FIFO/ADC interfaces, ingestible devices and motors in final sheets are not imported merely because the sheet includes them.

> **中文：** 筛选决定：Week8 p30起超出声明范围；一页纸出现的后续FIFO/ADC接口、吞服装置及电机不能仅因出现就纳入本次Quiz。

[D local details] Conformal-electrode technical design / 贴合电极技术设计

EN: Past overlap: past2 p2 has flexible-device applications; slide comparison

> **中文：** 历年重合：past2 p2 has flexible-device applications; slide comparison

EN: Current classroom evidence: W8 p14–17 | T8-0018–0020

> **中文：** 本年课堂依据：W8 p14–17 | T8-0018–0020

EN: Decision: The lecturer explicitly excludes these technical requirements. This corrects the previous practice bank’ s excessive emphasis on the conformal-design question.

> **中文：** 筛选决定：老师明确排除该技术要求；据此修正旧题库对贴合设计题的过度强调。

Boundaries that must remain qualified / 必须保留的边界

EN: Do not broadly exclude OCT, EMG, ultrasound or filters because one local formula/detail is secondary. W4’ s annual resolution statement targets the final; W5 supplies the separate Quiz promise. The lactate example does not clearly identify Quiz versus final. The RC-cutoff exclusion is locally phrased for this year’ s exam. No exact “no derivations” statement was found; use the actual short-calculation/long-formula guidance.

中文： 不能因为某个局部公式或细节次要，就排除整个OCT、EMG、超声或滤波主题。W4“每年分辨率题”针对期末，W5另有Quiz承诺。乳酸例题没有明确区分Quiz与期末；RC例题排除在局部语境中说“今年考试”。未找到逐字“无需推导”，应采用真实的短计算与长公式指导。

Corrections and unresolved sample details / 纠错与样题未决细节

EN: Sample Q10: corrected B=5. Sample Q5: approximately 45 Hz, reference C=40–48 Hz, with limited graph-reading precision. Sample Q9: B is a derived CCD/CMOS reference, while the spoken review hesitates rather than giving a clear final answer. Sample Q12: the intended numerical option is A≈0.55, but the current versus current-density units are inconsistent; do not hide this dimensional issue.

中文： 样题Q10：修正B=5。Q5：约45 Hz，参考C=40–48 Hz，读图精度有限。Q9：B来自CCD/CMOS原理推导；口头回顾有犹豫，没有明确最终字母。Q12：题意数值选项为A≈0.55，但电流与电流密度单位不一致，不能隐去量纲问题。

EN: Source-note corrections: the typed past sheet calls X-rays “non-ionizing”; that is wrong. X-rays are ionizing radiation. The Week 6 p83 inductor relation is printed incorrectly as 2πL/f; the correct ideal reactance is X_L=2πfL. Lifetime is a 1/e decay time, not automatically a half-life; depolarization can make the cell interior positive. These points are corrected rather than copied from notes or ASR.

中文： 源材料纠错：往年打字一页纸把X射线写成“non-ionizing”，错误；X射线属于电离辐射。Week6 p83把电感关系式印为2πL/f，正确理想电抗是X_L=2πfL。Lifetime是1/e衰减时间，不直接等于半衰期；去极化时细胞内可变正。这些内容已纠正，不照抄笔记或字幕识别错误。

EN: PMT correction: the lecture description at T4-0049–0050 wrongly describes tiny mirrors and optical reflection. The actual diagram (W4 p35) labels dynodes: multiplication is by secondary electron emission. Amplification does not remove all noise. See Hamamatsu’s PMT explanation.

中文： PMT纠错：T4-0049–0050把结构说成小镜子与光反射，讲解不准确。原图W4 p35标有dynodes倍增极，实际为二次电子发射倍增；放大不等于消除全部噪声。已核对 Hamamatsu官方PMT说明。

EN: The X-ray “non-ionizing” error also occurs in the original W4 p7, not only a student sheet. Correct it to ionizing; see FDA medical X-ray imaging. W7 p84 also misprints gamma as <25 Hz; learn the band ordering here rather than memorizing this inconsistent boundary.

中文： X射线“non-ionizing”的错误也出现在原课件W4 p7，不只是学生一页纸；应改为电离性，见FDA医学X射线说明。W7 p84还把gamma误写为<25 Hz；这里保留频带顺序，不让你死背不一致的边界。

EN: Further wording safeguards: the glucose/peroxide example is not universal to amperometry; high input impedance is not voltage gain; below-threshold neurons still have a resting potential; ECG leads are not the P/QRS/T waves. These conceptual corrections take precedence over loose oral wording.

中文： 另已防止四种过度概括：葡萄糖/过氧化氢例子不代表全部安培传感；高输入阻抗不等于高电压增益；未达阈值仍有静息电位；心电导联不等于P/QRS/T波。复习采用纠正后的概念，不照抄宽泛口头表述。